Kalyani Tekade

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EDUCATION

Master of Science in **Computer Science** May 2025
Arizona State University, Tempe, AZ 4.0/4.0
Relevant Courses: Data Mining, Statistical Machine Learning, Data Processing at Scale, Time Series Analysis, Multimedia and Web Databases
Bachelor of Engineering in Electronics and Communication May 2023
Ramaiah Institute of Technology, Bangalore, India 9.3/10

TECHNICAL SKILLS

Programming Language / Databases : Python, HTML/CSS, JavaScript, C, MySQL, PostgreSQL, MongoDB
Data Analysis and Visualization : Excel, Power BI, NumPy, Pandas, Matplotlib, Seaborn
Machine and Deep Learning Frameworks : TensorFlow, Sklearn, PyTorch, Keras, Torchvision, SciPy
Big Data Tools : Hadoop, Sqoop, Hive, Scala Programming, Spark, HBase, Cassandra, Kafka
DevOps : Terraform, Git, Ansible, Jenkins, Docker, Kubernetes, Jira
Certificate : Google Data Analytics , Microsoft Excel Expert

PROFESSIONAL EXPERIENCE

Unilever, Bangalore, Data Analyst January2023 – August 2023

- Implemented **ETL** processes for data cleaning and standardization on data points for the Unilever Technology Portfolio Dashboard using **Apache Spark**, resulting in improved data quality and accuracy for visualization in **Power BI**; achieved a 30% reduction in data inconsistencies.
- Automated the ETL pipeline using **Apache Kafka** for real-time data ingestion, incorporating stream processing and real-time analytics to ensure efficient data flow and timely updates for accurate visualization.
- Optimized data collection and transformation by designing an advanced **Excel** system featuring auto-fill functions and dropdown menus, increasing update frequency by 25%.

Centre for Artificial Intelligence & Robotics, Bangalore, Machine Learning Researcher January 2022 – December 2022

- Performed data wrangling and preprocessing for IMU Calibration using **SQL, Pandas,** and **Sklearn,** reducing anomalies by 95%. Built algorithms like SVMs, Random Forests, and DNNs using **TensorFlow** and **PyTorch,** increasing model accuracy by 20%.
- Prepared the entire ML pipeline, enhancing **Python** code efficiency and model scalability by 30% using a modular design.
- Streamlined the pipeline using **Terraform** and **Jenkins**, ensuring continuous integration and deployment (CI/CD) for seamless model updates and maintenance, reducing deployment time by 40%.

Govt. of India, Hyderabad, Data Science Intern June 2021 – December 2021

- Developed a **CNN** for image classification and enhanced interpretability using **LIME, SHAP,** and **EDA** by 30%. Evaluated CNN architectures like LeNet-5, ResNet-50, VGG, and YOLO, improving classification accuracy by 20%.
- Applied object detection, image classification, and semantic segmentation techniques, reducing error rates by 25%.
- Created detailed technical documentation, presentations, and visualization using **Microsoft Suite** (Word, PowerPoint, Excel), leading to a 15% reduction in development time for future projects.

PROJECTS

IntelliBot: Enhanced Chatbot Intent Recognition Link July 2024

- Deployed a chatbot using TensorFlow and NLP techniques, incorporating K-means clustering for labeling and classifying chat intents, resulting in a 25% improvement in clustering performance.
- Designed an evaluation framework using confusion matrices and classification reports, and established a robust training pipeline with Keras Tuner for hyperparameter optimization and TensorFlow Hub for deployment, resulting in a 30% improvement in scalability and performance.

LoyaltyBooster: Optimizing Customer Retention with Ensemble Models Link May 2024

- Executed Logistic, Random Forest, AdaBoost, and Gradient Boosting using Python libraries, resulting in a 20% increase in model accuracy and amplified customer retention predictions.
- Developed feature engineering and model validation techniques to enhance predictive performance, achieving a 15% improvement in recall and precision metrics, aiding in better identification of at-risk customers.

Resume Evaluator Link May 2024

- Designed resume evaluation approaches using Peer Comparisons and Deep Learning (Doc2Vec), achieving high accuracy in matching resumes to job descriptions, with the best model having a similarity score of 70.1%.
- Designed and deployed a web-based user interface using Flask that enables users to input job descriptions and resumes; the platform generates a real-time match score, streamlining evaluation.

SalesPredictor: Forecasting Daily Sales for Rossmann Stores Link December 2023

- Built a sales forecasting model for Rossmann stores by leveraging historical sales data and applying Linear, SGD, Random Forest, and Decision Tree Regression using Python, which led to a 20% decrease in forecast errors.
- Enhanced the data pipeline by integrating robust feature engineering techniques and employing cross-validation and hyperparameter tuning, resulting in a 25% increase in model precision and providing more reliable sales forecasts.

RideForecastor: Predictive Modeling for Ride Request Demand Link December 2023

- Engineered a demand prediction model for ride requests using Linear, Decision Trees, Random Forest, and Gradient Boosting Regression, resulting in a 30% improvement in prediction accuracy, optimizing ride allocation, and reducing wait times.
- Implemented a prediction pipeline with modularized code for efficient model training and evaluation, achieving a 20% increase in development speed and facilitating scalable deployment for real-time ride demand forecasting.

WORK EXPERIENCE

Arizona State University, Graduate Service Assistant January 2024 – Present

- Graded assignments and projects for CSE 572 (Data Mining), providing feedback to 150+ students to enhance understanding of data mining concepts.